Chapter 24

Direct Gold Restorations

"The most pitiful among men is he who turns his dreams into silver and gold…".

—Khalil Gibran

Direct Gold

Direct gold is a gold restorative material that is manufactured for compaction directly into prepared cavities. Two types of direct gold are manufactured for dental use: gold foil and powdered gold. These gold materials differ in their metallurgic structure.

Pure gold has been in use in dentistry for more than 100 years.[1–6] It is generally agreed that this noble metal is a *superior restorative material* for treatment of many small lesions and defects in teeth, given sound pulpal and periodontal health. Direct gold restorations can last for a *lifetime* if attention is paid to details of restorative technique and to proper home care. The longevity of direct gold restorations is a result of the *superb biocompatibility* of gold with the oral environment and its *excellent marginal integrity*.

Definition. Direct gold is a gold restorative material that is manufactured for compaction directly into prepared cavities.

Indications

I. Class I direct gold restorations are one option for the treatment of *small carious lesions in pits* and *fissures* of most posterior teeth and the lingual surfaces of anterior teeth.
II. Direct gold also is indicated for treatment of small, cavitated *Class V carious lesions* or for the restoration, when indicated, of abraded, eroded or abfraction areas on the facial surfaces of teeth (although access to the molars is a limiting factor).
III. Class III direct gold restorations can be used on the *proximal surfaces of anterior teeth* where the lesions are small enough to be treated with esthetically pleasing results.
IV. Class II direct gold restorations are an option for restoration of *small cavitated proximal surface carious lesions* in posterior teeth in which marginal ridges are not subjected to heavy occlusal forces (e.g. the mesial or distal surfaces of mandibular first premolars and the mesial surface of some maxillary premolars).
V. Class VI direct gold restorations may be used on the incisal edges or cusp tips.
VI. A defective margin of an otherwise acceptable cast gold restoration also may be repaired with direct gold.

Contraindications

Direct gold restorations are contraindicated in some patients whose teeth have:

I. Very large pulp chambers
II. Severely periodontally weakened teeth with questionable prognosis
III. Patients for whom economics is a severely limiting factor
IV. In handicapped patients who are unable to sit for the long dental appointments required for this procedure
V. Root canal-filled teeth are generally not restored with direct gold

Types of Direct Gold

Several physical types of direct-filling gold have been produced.[7] All are *compactable* in that they are inserted into tooth preparations under force and compacted or condensed into the prepared line and point angles and against the walls. Two types of direct gold are manufactured for dental use and these gold materials differ in their metallurgic structure:

I. Gold foil.
II. Powdered gold.

Gold Foil

I. Gold foil is manufactured by beating pure gold into thin sheets.

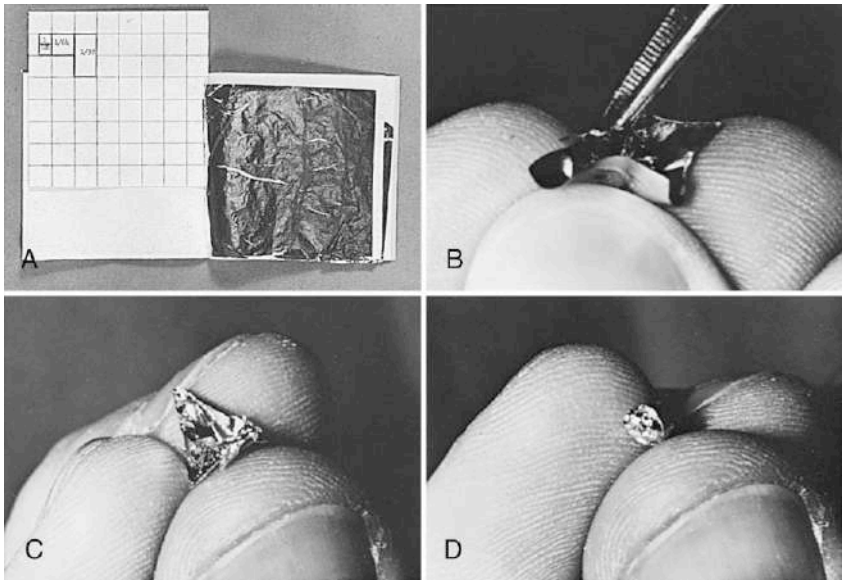


Figure 24.1 (A) 4 × 4 in. book of foil marked for cutting and rolling into pellets of various sizes. (B and C) Corners of foil piece are tucked into centre. (D) Foil is rolled into a completed pellet.
(Part A, Courtesy: Terkla and Cantwell.)

II. The gold foil is cut into 4 × 4 in. (10 × 10 cm) sheets and sold in books of sheets, separated by pages of thin paper. The books contain 1/10 oz or 1/20 oz of gold.
III. The sheet of foil that weighs 4 g is termed *No. 4 foil*; the sheet weighing 3 g is termed *No. 3 foil*; and the sheet weighing 2 g is termed *No. 2 foil*.
IV. Because the 4 × 4 in. sheets are too large to be used in restorative procedures, they are rolled into cylinders or *pellets* before insertion into tooth preparations.
V. Pellets of gold foil are generally rolled from 1/32 in., 1/43 in., 1/64 in. or 1/128 in. sections cut from a No. 4 sheet of foil. The book of foil is marked and cut into squares or rectangles (Fig. 24.1A).
VI. Each piece is placed on clean fingertips, and the corners are tucked into the centre (Fig. 24.1B and C), and then the foil is lightly rolled into pellet form (Fig. 24.1D).
VII. After pellets of gold are rolled, they may be conveniently stored in a gold foil box (Fig. 24.2), which is divided into labelled sections for various sizes of pellets.
VIII. Preferential contamination is suggested by placing a damp cotton pellet dipped in *18% ammonia* into each section of the box. This serves to prevent deleterious oxides from forming on the gold until it is used.

Powdered Gold

I. Powdered gold is made by a combination of chemical precipitation and atomization (Fig. 24.3A).[8]



Figure 24.2 Gold foil box. Compartments are labelled to show pellet size.

II. The atomized particles are mixed together in wax, cut into pieces and wrapped in No. 4 or No. 3 foil (Fig. 24.3B).
III. Several sizes of these pellets are available. This product is marketed as *Williams E-Z Gold* (Ivoclar-Williams, Amherst, NY, USA).

Principles of Manipulation of Direct Gold

The ability of gold to be used as a direct restorative material is explained in Box 24.1:

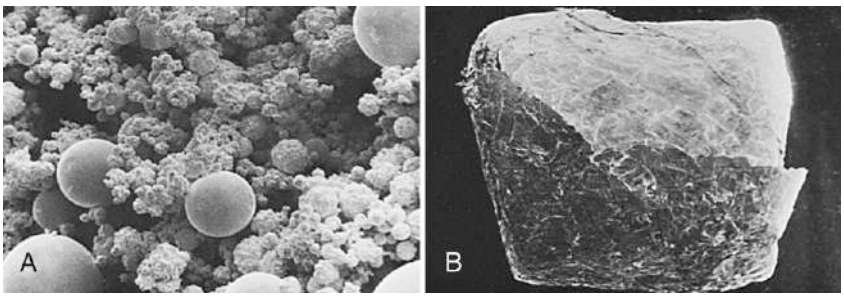


Figure 24.3 Scanning electron micrographs of direct-filling gold. (A) Spheres of E-Z Gold. (B) Wrapped E-Z pellet that contains spheres. (Courtesy: Ivoclar-Williams Company, Inc., Amherst, NY.)

Box 24.1 Principles behind compaction of direct gold

Direct gold that is in a *cohesive state* should only be used

↓

Direct gold is inserted into tooth preparations and a suitable, biologically compatible *compacting force* must be delivered under force

↓

Welding occurs because pure gold with an absolutely clean surface coheres as a result of metallic bonding

↓

The purpose of the force is to *weld* the gold into restorations containing *minimal porosity* or internal void spaces[9–11]

↓

As the gold is forced and compressed into a tooth preparation, *succeeding increments* cohere to those previously placed

Cohesion and Degassing

Cohesive and Noncohesive Gold

Direct gold may be either cohesive or noncohesive. The manufacturer supplies books of gold foil or prerolled cylinders in a cohesive or noncohesive state.

I. **Cohesive gold**: The gold is considered to be *cohesive* in nature when its surface is pure and free of surface oxides and impurities.
II. **Noncohesive gold:** It is considered *noncohesive* in the presence of *surface impurities or wax*, which prevents one increment of gold from cohering to another.

Degassing

Definition

Gold attracts gases that render it noncohesive, and the process of removal of such gases from the surface of the gold by application of heat is referred to as *degassing* or *annealing*.

Technique

Degassing is accomplished by heating the gold foil by one of the following three methods (Fig. 24.4):

I. Mica tray over a flame
II. Electric annealer
III. Pure ethanol flame

Degassing gold foil

Any of the three degassing procedures is satisfactory for gold foil. However, the advantage of the technique involving the use of the pure ethanol flame is that each piece of gold is selected and heated just before insertion, and waste of gold is avoided. The gold is passed into the blue inner core of the flame on the tip of a foil-passing instrument and held just until the gold becomes dull red, and then the instrument is withdrawn from the flame. After a few seconds are allowed for cooling, the gold is placed in the preparation.

Degassing powdered gold

The only technique for powdered gold degassing is employing the pure ethanol flame. The E-Z Gold pellet must be heated 0.5–1 in. above the ethanol flame until a bright flame occurs (caused by ignition of the wax) and the pellet becomes dull red for 2–3 s, then it is withdrawn from above the flame.

I. All direct-filling gold products are degassed *immediately before use* except when noncohesive foil is specifically desired.
II. *Underheating* during degassing should be avoided because it fails to render the gold surface pure.
III. *Overheating* also should be avoided because it may cause the gold to become brittle or melt and render it unusable.

Compaction of Direct Gold

Principles

Direct-filling gold must be compacted during insertion into tooth preparations.[12] Two fundamental principles involved in compaction of cohesive gold are:

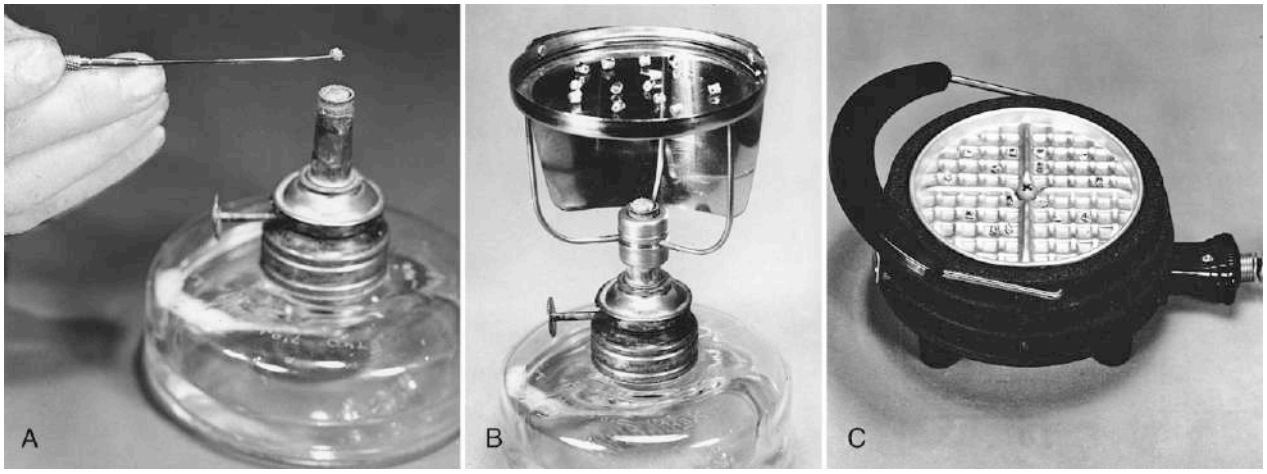


Figure 24.4 (A) Pellet of gold foil is degassed in pure ethanol flame. (B) Mica tray mounted over alcohol lamp for degassing several increments of gold simultaneously. (C) Gold foil degassed on an electric annealer.
(Courtesy: Terkla and Cantwell.)

I. *Weld* the gold into a cohesive mass.
II. *Wedge* as much gold as possible into the tooth preparation.[13]

Welding takes place primarily as a result of the coherence of the noble metal to itself. Wedging results from careful compacting technique. Regardless of the technique used, there would be void spaces not only in the compacted gold but also along the preparation walls. Success depends on *minimizing these voids*, particularly on the surface of the restoration and at the cavosurface interface, where leakage to the internal aspects of the restoration may begin.

Methods of Compaction

The following are the various methods available for compacting direct gold:

I. Compaction with mallet and condensers
   i. Hand mallet (Fig. 24.5A)
   ii. Electro-mallet
   iii. Pneumatic mallet
II. Compaction with hand condensers
   i. With the exception of E-Z Gold, the compaction takes the form of malleting forces that are delivered either by a hand mallet used by the assistant or by an electro-mallet or a pneumatic mallet used by the dentist.
   ii. E-Z Gold, because of its powdered form, may be compacted by heavy hand pressure delivered in a rocking motion with specially designed hand condensers.[14,15]

Condensers for Compaction

I. Condensers are designed to deliver forces of compaction to direct gold (Fig. 24.5B).
II. Condensers used in the *handpieces* of the electro-mallet or pneumatic mallet consist of a *nib*, or working tip, and a short shank that fits into the malleting handpiece.
III. Condensers used with the *hand mallet* are longer and have a blunt-ended handle that receives light blows from the hand mallet.
IV. Condenser nibs are available in several shapes and sizes and have *pyramidal serrations* on the nib faces to prevent slipping on the gold.

Compaction Technique for Gold Foil

Stepping Motion

Compaction begins when a piece of gold is placed in a tooth preparation. The gold is first pressed into place by hand, then a condenser of suitable size is used to begin malleting in the centre of the mass (often this is done while this first increment is held in position with a holding instrument). Each succeeding step of the condenser overlaps (by half) the previous one as the condenser is moved towards the periphery in an orderly *stepping motion* (Fig. 24.6). The gold moves under the nib face of the condenser, effecting compaction as malleting proceeds.

Line of Force

The line of force is important when any gold is compacted. The line of force is the direction through which the force is delivered (i.e. the direction in which the condenser is aimed) (Fig. 24.7). Research has shown that a biologically acceptable pulpal response occurs after proper direct gold procedures.[16] Correct hand-malleting technique requires a light, bouncing application of the mallet to the condenser, rather than delivery of heavy blows.

Compaction Technique for E-Z Gold

Using an amalgam condenser or a gold foil condenser, the first pellet of E-Z Gold is pressed into the depth of the

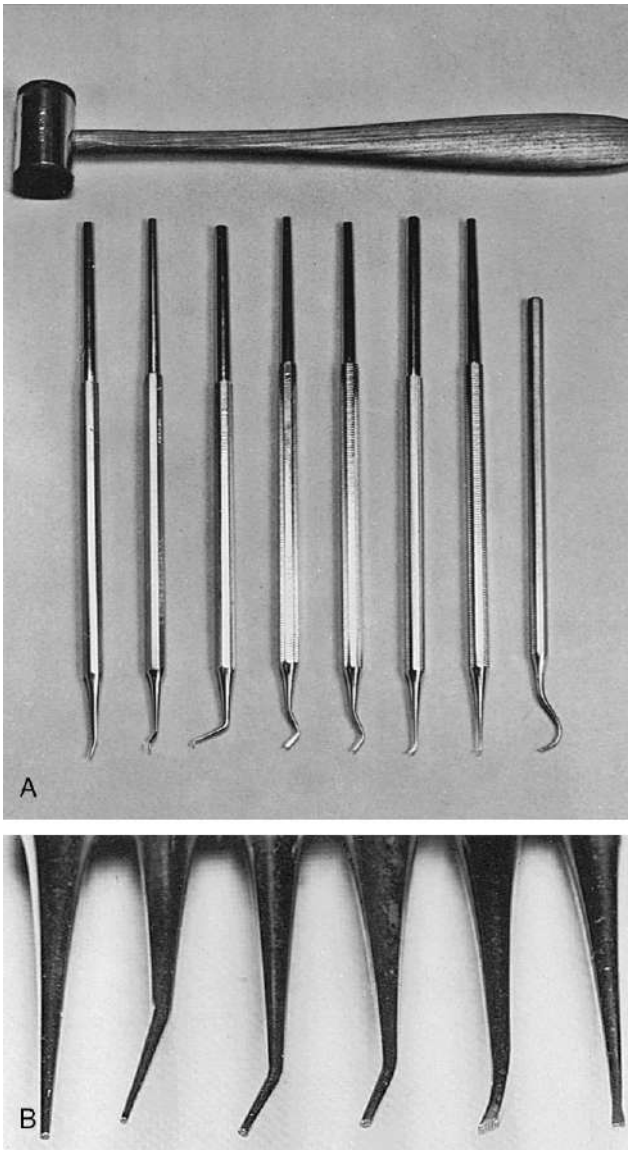


Figure 24.5 (A) Hand mallet and condensers used for hand mallet compaction of direct gold. (B) Selection of variously shaped nibs. *Left to right*, Three round-faced nibs, oblique-faced nib, foot condenser and rounded rectangular nib.
(Part A, Courtesy: Terkla and Cantwell.)

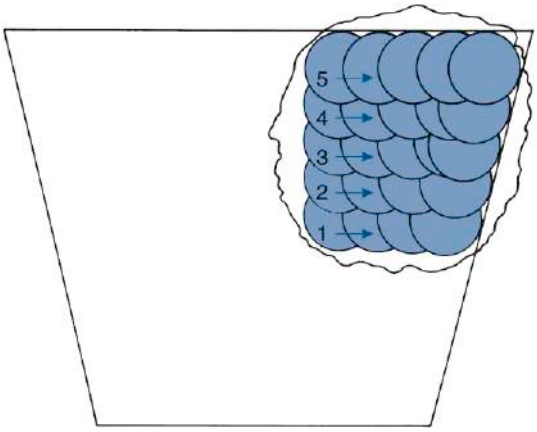


Figure 24.6 Diagrammatic order of compaction for increment of direct-filling gold. Condensers are moved across surface of gold in an orderly stepping motion. Each succeeding step of the nib overlaps the previous one by at least half of the nib face diameter. Condensation begins at position 1 and moves to the right, then resumes at 2 and repeats movement to the right. Finally, it continues in rows 3, 4 and 5.

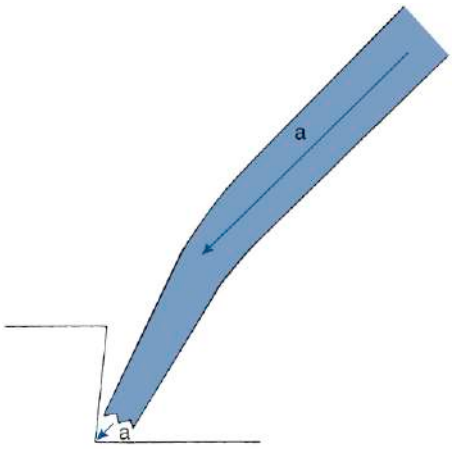


Figure 24.7 Line of force (*a*) remains parallel with the shaft or handle of the condenser, regardless of any angles in the shank of the instrument.

tooth preparation and tamped into position. A small condenser is selected to thrust and wedge the gold into opposing line angles and against opposing walls, to secure the mass in the preparation. Additional pellets are added (one at a time, banking against the prepared walls) until the entire preparation is filled.

Principles of Tooth Preparation for Direct Gold Restorations

The principles of tooth preparation for all direct gold restorations demand meticulous attention to detail for success. The step wise tooth preparation and restoration of a Class I lesion with direct gold is shown in Figs. 24.8–24.14.

Outline Form

I. The margins in outline form must not be ragged.
II. They are established on sound areas of the tooth that can be finished and polished.
III. The outline must include all structural defects associated with the lesion.
IV. The marginal outline must be designed to be esthetically pleasing because the final restoration may be visible.

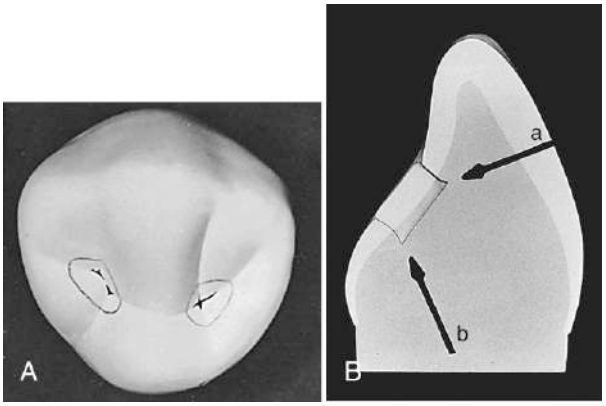


Figure 24.8 (A) Typical Class I occlusal marginal outlines for pit restorations with direct gold. (B) Cross-section of model of lingual Class I preparation on maxillary incisor. Undercuts (*a* and *b*) are placed in dentin incisally and gingivally for additional retention.

Failure to give attention to outline form may result in an unsightly restoration or, at the least, one in which cavosurface deficiencies are immediately obvious.

Resistance Form

I. Resistance form is established by orienting the pulpal wall to be flat and perpendicular to the occlusal forces.
II. All enamel must be supported by sound dentin.
III. Optimally placed axial or pulpal walls promote the integrity of the restored tooth, providing a suitable thickness of remaining dentin.

Poor resistance form can result in tooth fracture; inadequate retention form may result in a loose restoration that is frustrating to the dentist.

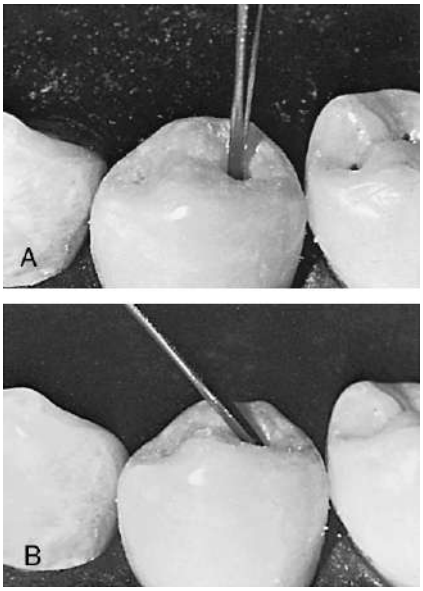


Figure 24.10 (A) Compaction forces are delivered by the condenser held at 90 degree angle to the pulpal wall. (B) Gold is condensed against the external preparation walls.

Retention Form

I. The retention form is established by parallelism of some walls and by strategically placed converging walls.
II. In addition, walls must be smooth and flat.
III. Internal line angles must be *sharp* (to resist movement).
IV. Internal form includes an initial depth into dentin, ranging from 0.5 mm from the dentinoenamel junction (DEJ) in Class I preparations to 0.75 mm from cementum in Class V preparations.

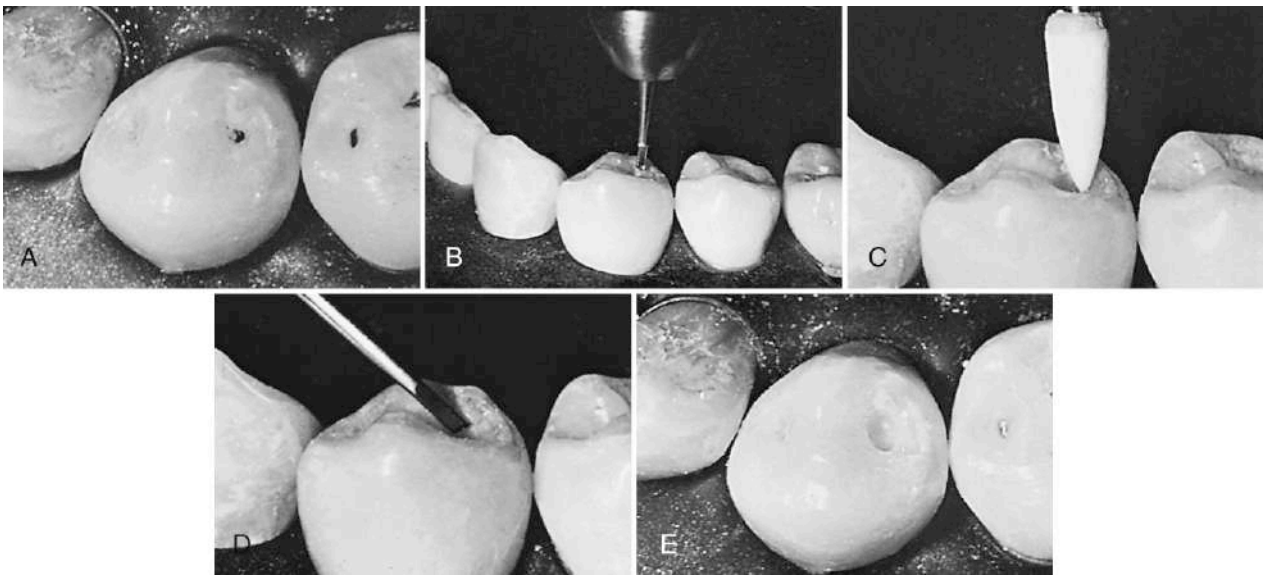


Figure 24.9 Class I preparation for direct gold. (A) Preoperative view of pit lesion. (B) No. 330 bur is aligned properly for occlusal preparation. (C) Occlusal cavosurface bevel is prepared with white stone. (D) The bevel may be placed with an angle former. (E) Completed tooth preparation.

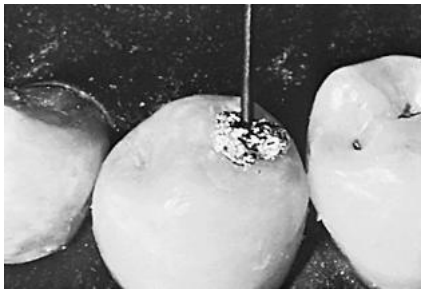


Figure 24.11 Placement of pellet of gold foil and compaction into tooth preparation.

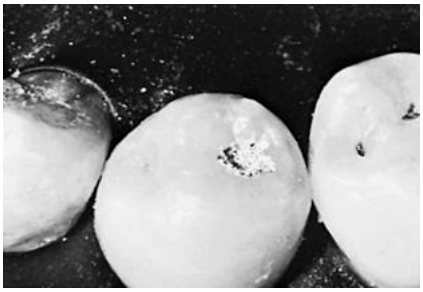


Figure 24.12 Compaction of gold foil has proceeded sufficiently to cover all the cavosurface margins.

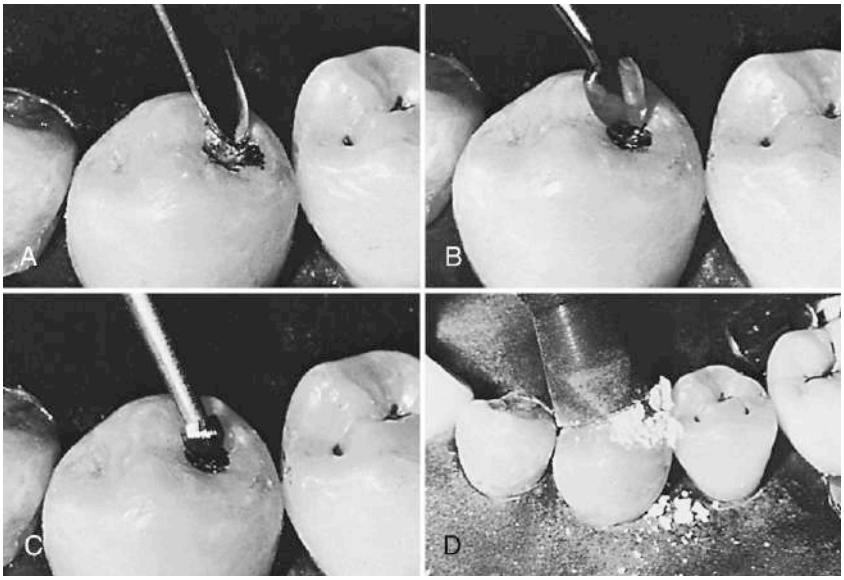


Figure 24.13 Steps in finishing Class I direct gold restoration. (A) Burnisher work-hardens the surface gold. (B) The cleoid–discoid instrument removes the excess gold from the cavosurface margins. (C) A No. 9004 bur is used to begin the polishing phase. (D) Polishing abrasives are applied with a rubber cup.



Figure 24.14 Completed restoration.

Failure to keep the internal line angles would make the restoration to fail due to poor retention form.

Convenience Form

I. Optimal convenience form requires suitable access and a dry field provided by the rubber dam.

II. Access additionally may require the use of a gingival retractor for Class V restorations or a separator to provide a minimal amount of separation (0.5 mm maximum) between anterior teeth for Class III restorations.

III. Removal of remaining carious dentin, final planning of cavosurface margins and debridement complete the tooth preparation for direct gold.

Lack of detailed convenience form may render an otherwise excellent tooth preparation unrestorable.

Summary

Direct-filling gold is useful in restorative dentistry. If carefully manipulated by a dentist, this restorative material may provide lifetime service to patients and promote their oral health. Direct-filling gold contributes to the art and the science of restorative dentistry.

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